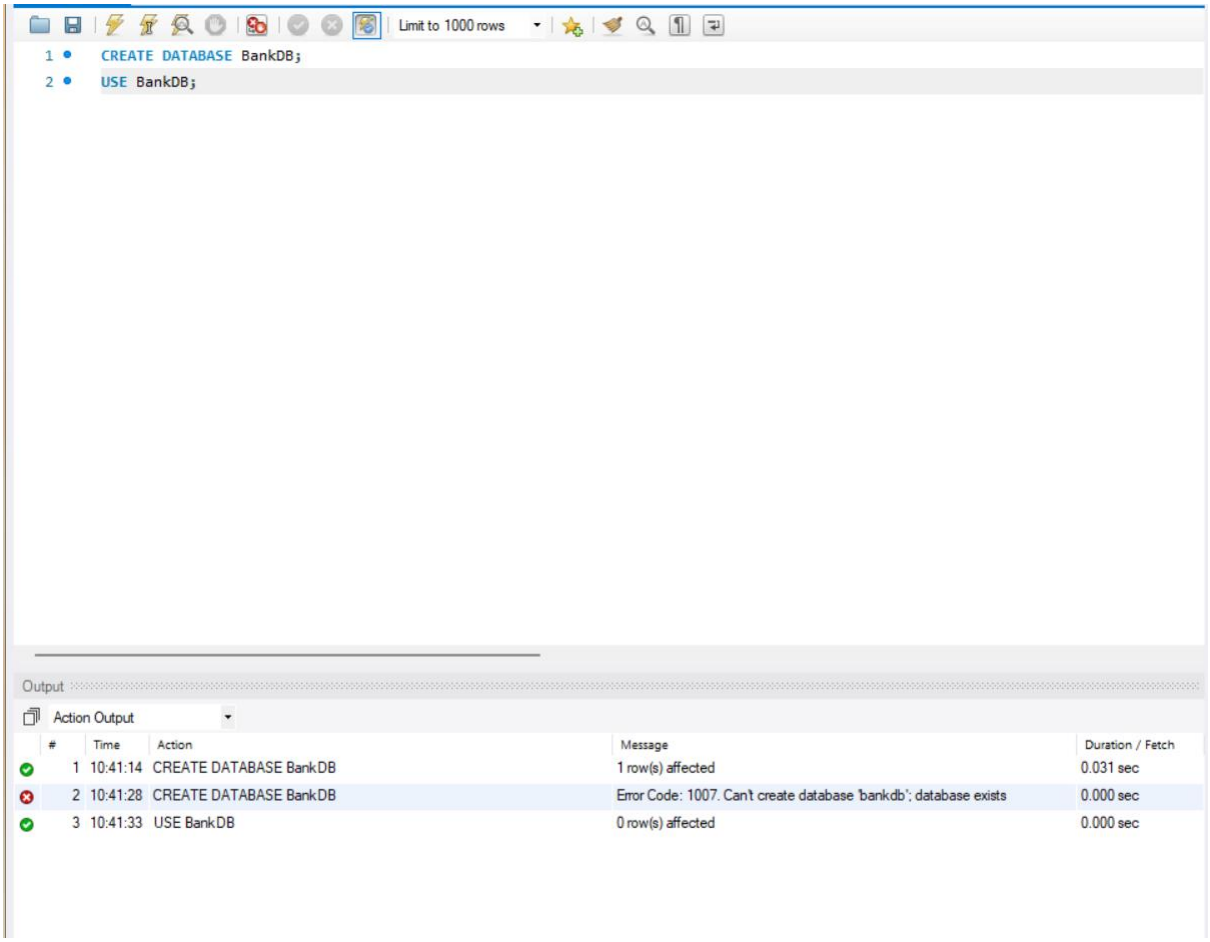
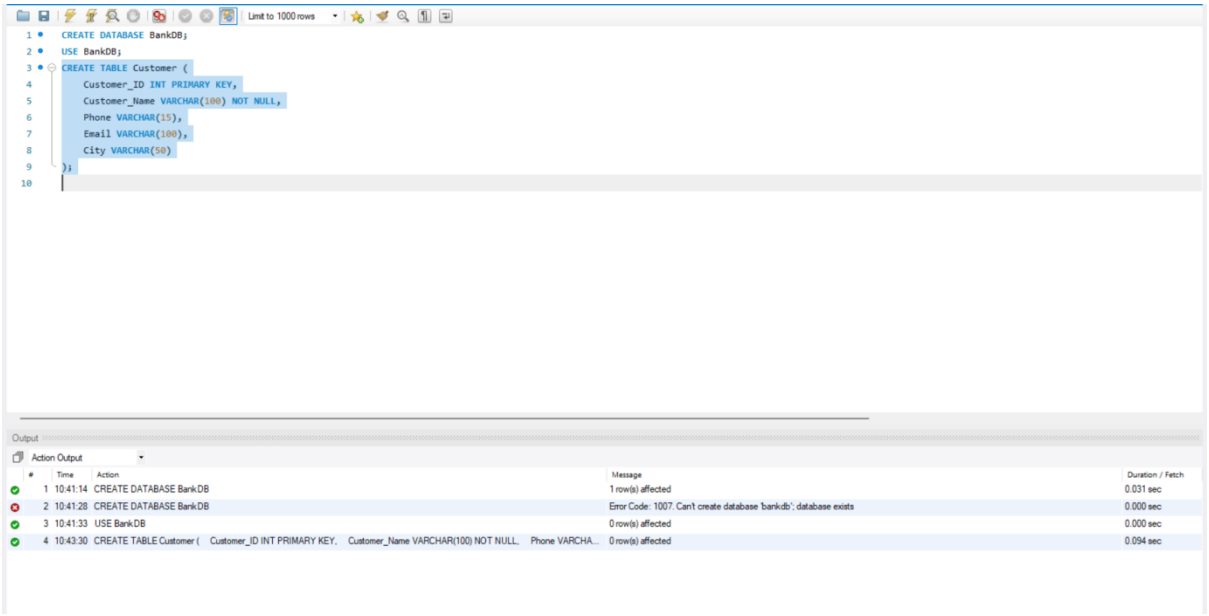
1.CREATE DATABASE



2. CREATE CUSTOMER TABLE



3.CREATE ACCOUNT TABLE

```sql
CREATE DATABASE BankDB;
USE BankDB;
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2) DEFAULT 0,
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:41:14	CREATE DATABASE BankDB	1 row(s) affected	0.031 sec
2	10:41:28	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
3	10:41:33	USE BankDB	0 row(s) affected	0.000 sec
4	10:43:30	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.094 sec
5	10:44:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.094 sec

4. CREATE TRANSACTION TABLE

```sql
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);

CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:41:14	CREATE DATABASE BankDB	1 row(s) affected	0.031 sec
2	10:41:28	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
3	10:41:33	USE BankDB	0 row(s) affected	0.000 sec
4	10:43:30	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.094 sec
5	10:44:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.094 sec
6	10:45:13	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction...	0 row(s) affected	0.093 sec

5. CREATE LOAN TABLE

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

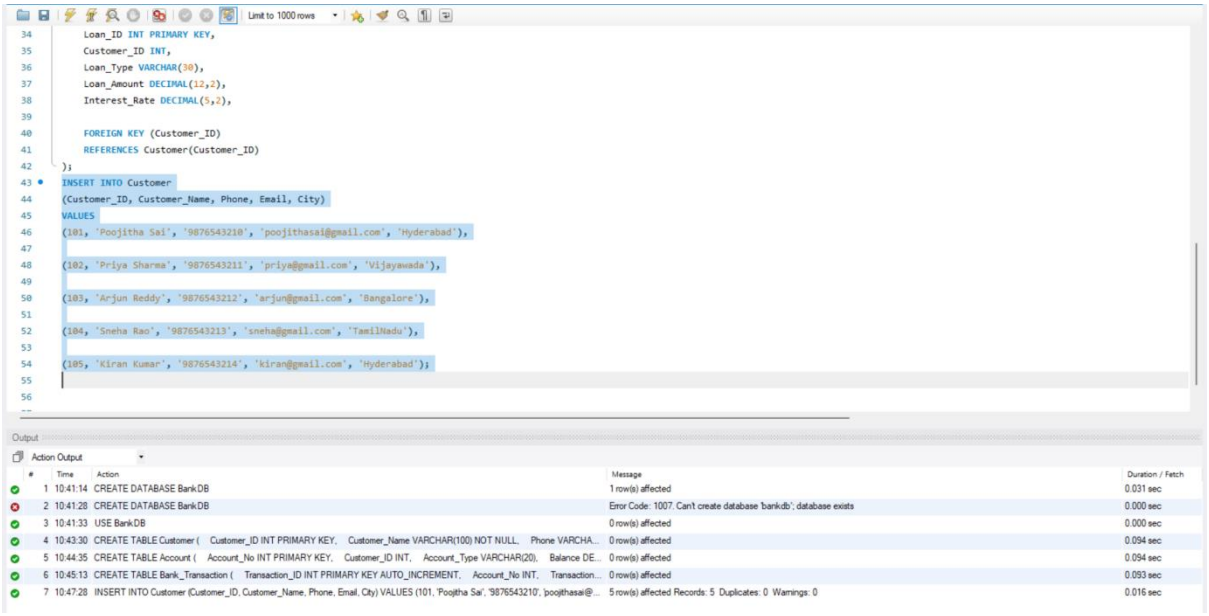
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:41:14	CREATE DATABASE BankDB	1 row(s) affected	0.031 sec
2	10:41:28	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
3	10:41:33	USE BankDB	0 row(s) affected	0.000 sec
4	10:43:30	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.094 sec
5	10:44:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.094 sec
6	10:45:13	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction...	0 row(s) affected	0.093 sec

6. INSERT CUSTOMER DATA



```
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Poojitha Sai', '9876543210', 'poojithasai@gmail.com', 'Hyderabad'),

(102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),

(103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),

(104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'TamilNadu'),

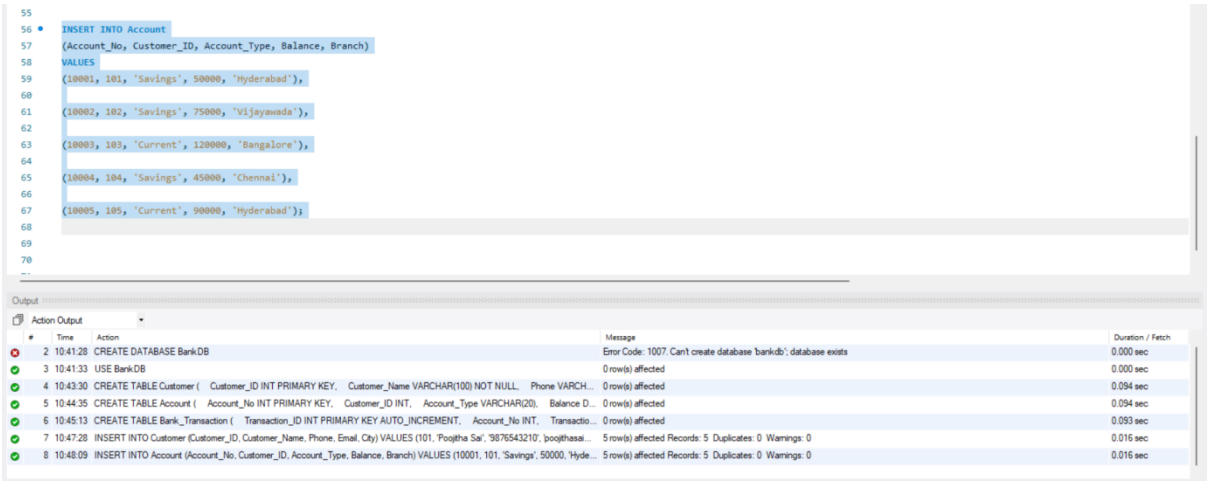
(105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:41:14	CREATE DATABASE BankDB	1 row(s) affected	0.031 sec
2	10:41:28	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
3	10:41:33	USE BankDB	0 row(s) affected	0.000 sec
4	10:43:30	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.094 sec
5	10:44:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.094 sec
6	10:45:13	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction...	0 row(s) affected	0.093 sec
7	10:47:28	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Poojitha Sai', '9876543210', 'poojithasai@...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

7. INSERT ACCOUNT DATA.



```
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

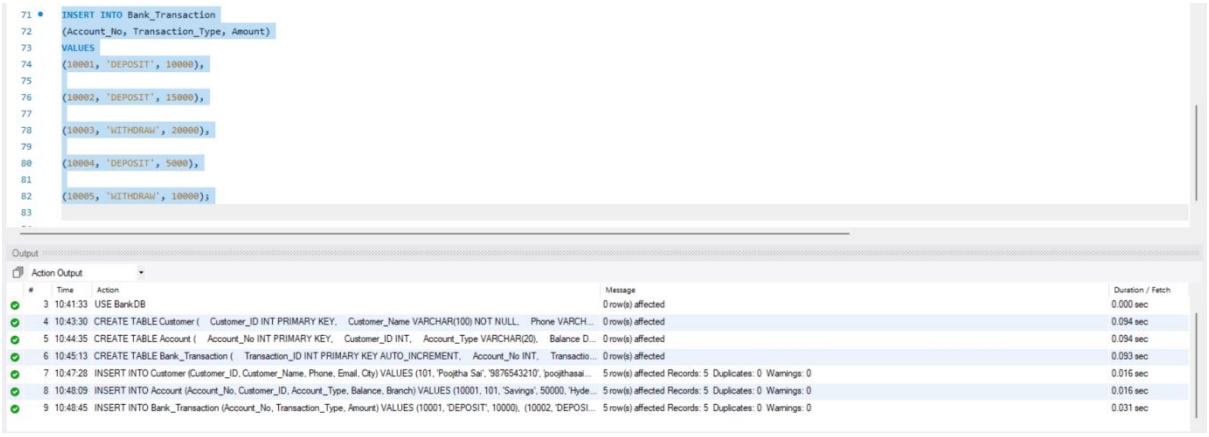
(10005, 105, 'Current', 90000, 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	10:41:28	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
3	10:41:33	USE BankDB	0 row(s) affected	0.000 sec
4	10:43:30	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCH...	0 row(s) affected	0.094 sec
5	10:44:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance D...	0 row(s) affected	0.094 sec
6	10:45:13	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transactio...	0 row(s) affected	0.093 sec
7	10:47:28	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Poojitha Sai', '9876543210', 'poojithasai...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
8	10:48:09	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyde...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

8. INSERT TRANSACTION DATA



```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	10:41:33	USE BankDB	0 row(s) affected	0.000 sec
4	10:43:30	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCH...	0 row(s) affected	0.094 sec
5	10:44:35	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance D...	0 row(s) affected	0.094 sec
6	10:45:13	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transactio...	0 row(s) affected	0.093 sec
7	10:47:28	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Poojitha Sai', '9876543210', 'poojithasai...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
8	10:48:09	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyde...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	10:48:45	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOSI...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.031 sec

9. INSERT LOAN DATA

```sql
USE BankDB;
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),

(504, 104, 'Personal Loan', 500000, 10.5);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:52:37	USE BankDB	0 row(s) affected	0.000 sec
2	10:52:45	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIMA...	0 row(s) affected	0.125 sec
3	10:53:01	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Home Loan', 5000000, 7....	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.047 sec

10. DISPLAY TABLE DATA.

```sql
Limit to 1000 rows
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),

(504, 104, 'Personal Loan', 500000, 10.5);

SELECT * FROM Customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Poojitha Sai	9876543210	poojithasai@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	TamilNadu
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

Customer 2

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:52:37	USE BankDB	0 row(s) affected	0.000 sec
2	10:52:45	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIM...	0 row(s) affected	0.125 sec
3	10:53:01	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Home Loan', 5000000, 7...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.047 sec
4	10:53:47	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

```sql
SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

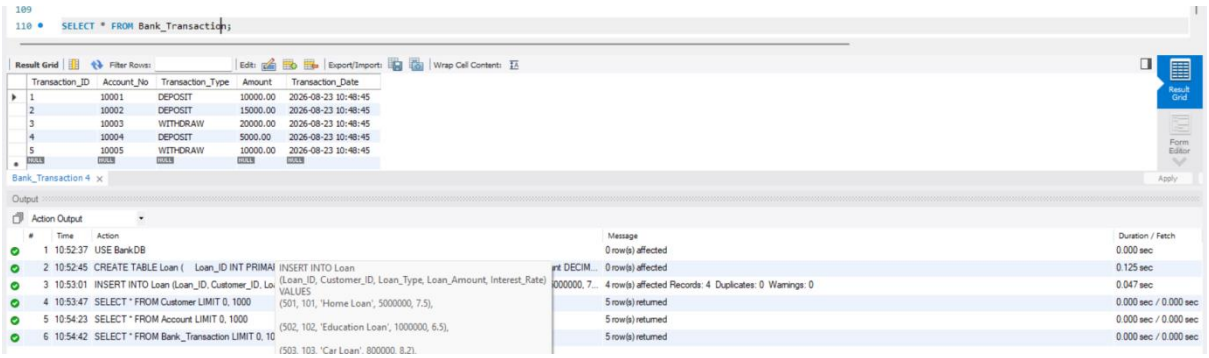
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 3

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:52:37	USE BankDB	0 row(s) affected	0.000 sec
2	10:52:45	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIM...	0 row(s) affected	0.125 sec
3	10:53:01	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Home Loan', 5000000, 7...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.047 sec
4	10:53:47	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	10:54:23	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec



```
SELECT * FROM Bank_Transaction;
```

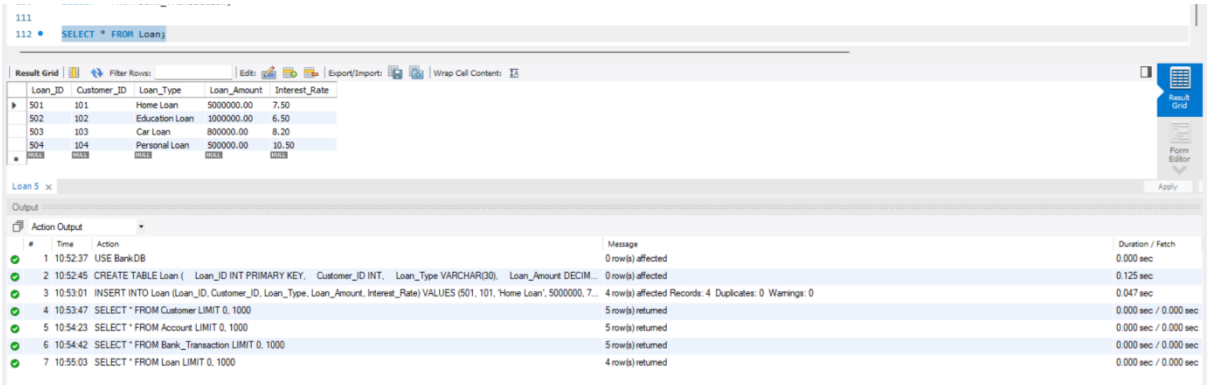
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-23 10:48:45
2	10002	DEPOSIT	15000.00	2026-08-23 10:48:45
3	10003	WITHDRAW	20000.00	2026-08-23 10:48:45
4	10004	DEPOSIT	5000.00	2026-08-23 10:48:45
5	10005	WITHDRAW	10000.00	2026-08-23 10:48:45

Bank_Transaction 4

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:52:37	USE BankDB	0 row(s) affected	0.000 sec
2	10:52:45	CREATE TABLE Loan (Loan_ID INT PRIMA	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Home Loan', 5000000, 7.5), (502, 102, 'Education Loan', 1000000, 6.5), (503, 103, 'Car Loan', 800000, 8.2),	0.125 sec
3	10:53:01	INSERT INTO Loan (Loan_ID, Customer_ID, Lo	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.047 sec
4	10:53:47	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	10:54:23	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
6	10:54:42	SELECT * FROM Bank_Transaction LIMIT 0, 10	5 row(s) returned	0.000 sec / 0.000 sec



```
SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

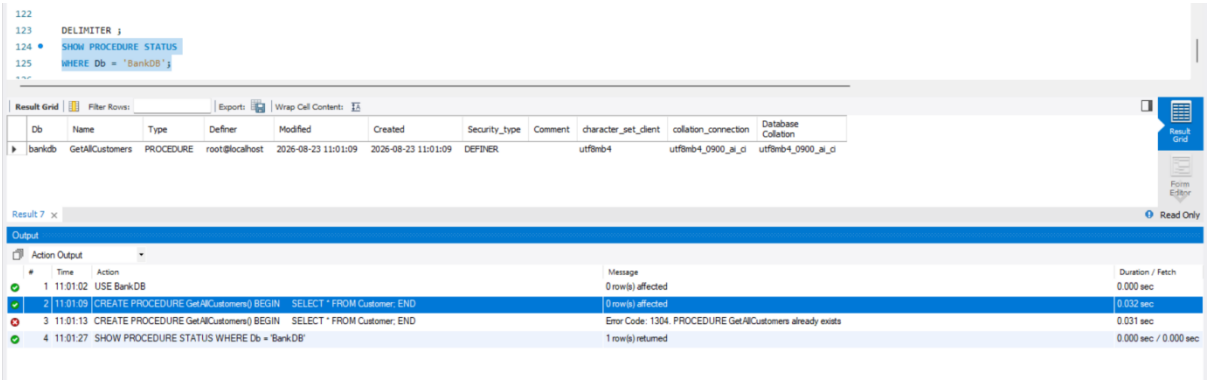
Loan 5

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:52:37	USE BankDB	0 row(s) affected	0.000 sec
2	10:52:45	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIM...	0 row(s) affected	0.125 sec
3	10:53:01	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Home Loan', 5000000, 7...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.047 sec
4	10:53:47	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
5	10:54:23	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
6	10:54:42	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
7	10:55:03	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

11. PROCEDURE TO DISPLAY ALL CUSTOMERS



```
DELIMITER ;
SHOW PROCEDURE STATUS
WHERE Db = 'BankDB';
```

Db	Name	Type	Definer	Modified	Created	Security_type	Comment	character_set_client	collation_connection	Database Collation
bankdb	GetAllCustomers	PROCEDURE	root@localhost	2026-08-23 11:01:09	2026-08-23 11:01:09	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8mb4_0900_ai_ci

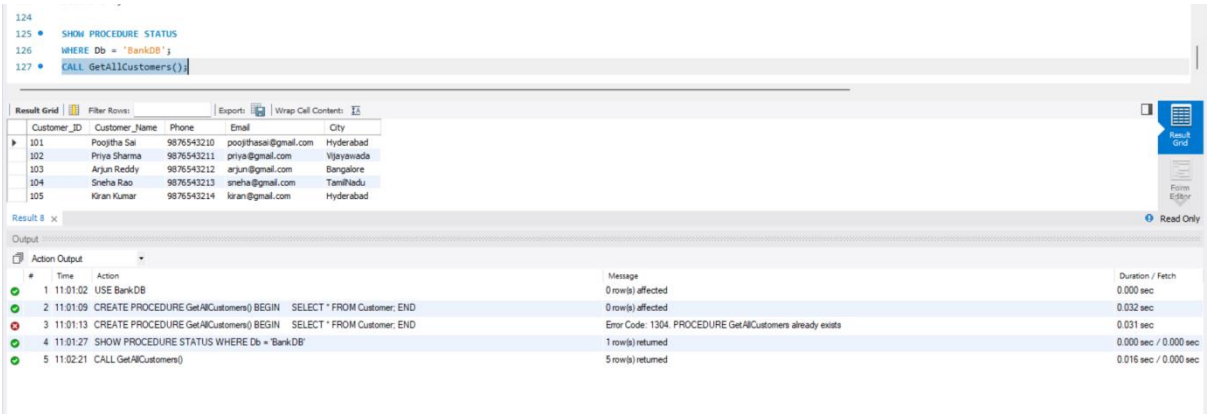
Result 7

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:01:02	USE BankDB	0 row(s) affected	0.000 sec
2	11:01:09	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected	0.032 sec
3	11:01:13	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	Error Code: 1304. PROCEDURE GetAllCustomers already exists	0.031 sec
4	11:01:27	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	1 row(s) returned	0.000 sec / 0.000 sec



```
SHOW PROCEDURE STATUS
WHERE Db = 'BankDB';
CALL GetAllCustomers();
```

Customer_ID	Customer_Name	Phone	Email	City
101	Poojitha Sai	9876543210	poojithasai@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	TamilNadu
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Result 8

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:01:02	USE BankDB	0 row(s) affected	0.000 sec
2	11:01:09	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected	0.032 sec
3	11:01:13	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	Error Code: 1304. PROCEDURE GetAllCustomers already exists	0.031 sec
4	11:01:27	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	1 row(s) returned	0.000 sec / 0.000 sec
5	11:02:21	CALL GetAllCustomers()	5 row(s) returned	0.016 sec / 0.000 sec

12. PROCEDURE TO DISPLAY ACCOUNT DETAILS

FOR A GIVEN ACCOUNT NUMBER

```
131
132 • CREATE PROCEDURE GetAccountDetails(
133       IN p_Account_No INT
134   )
135   BEGIN
136       SELECT *
137       FROM Account
138       WHERE Account_No = p_Account_No;
139   END //
140
141   DELIMITER ;
142
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:01:02	USE BankDB	0 row(s) affected	0.000 sec
2	11:01:09	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected	0.032 sec
3	11:01:13	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	Error Code: 1304. PROCEDURE GetAllCustomers already exists	0.031 sec
4	11:01:27	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	1 row(s) returned	0.000 sec / 0.000 sec
5	11:02:21	CALL GetAllCustomers()	5 row(s) returned	0.016 sec / 0.000 sec
6	11:03:17	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Account_...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right...	0.031 sec
7	11:04:05	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Account_...	0 row(s) affected	0.047 sec

```
142 • CALL GetAccountDetails(10001);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

Result 9 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	11:01:09	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected	0.032 sec
3	11:01:13	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	Error Code: 1304. PROCEDURE GetAllCustomers already exists	0.031 sec
4	11:01:27	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	1 row(s) returned	0.000 sec / 0.000 sec
5	11:02:21	CALL GetAllCustomers()	5 row(s) returned	0.016 sec / 0.000 sec
6	11:03:17	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Accoun...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the rig...	0.031 sec
7	11:04:05	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Account...	0 row(s) affected	0.047 sec
8	11:04:36	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec

13.PROCEDURE TO FIND CUSTOMER ACCOUNTS

```
164 • CALL GetCustomerAccounts(101);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Poojitha Sai	10001	Savings	50000.00	Hyderabad

Result 10 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	11:01:27	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	1 row(s) returned	0.000 sec / 0.000 sec
5	11:02:21	CALL GetAllCustomers()	5 row(s) returned	0.016 sec / 0.000 sec
6	11:03:17	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Accoun...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the rig...	0.031 sec
7	11:04:05	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Account...	0 row(s) affected	0.047 sec
8	11:04:36	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
9	11:06:03	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Customer_ID, C.Custom...	0 row(s) affected	0.047 sec
10	11:06:16	CALL GetCustomerAccounts(101)	1 row(s) returned	0.031 sec / 0.000 sec

14. PROCEDURE TO DEPOSIT MONEY

```
166     DELIMITER //
167
168 •   CREATE PROCEDURE DepositMoney(
169         IN p_Account_No INT,
170         IN p_Amount DECIMAL(12,2)
171     )
172     BEGIN
173
174         UPDATE Account
175         SET Balance = Balance + p_Amount
176         WHERE Account_No = p_Account_No;
177
178     END //
179
180     DELIMITER ;
181
182
183     CALL DepositMoney(10001, 5000);
184
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	11:03:17	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Accoun...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the rig...	0.031 sec
7	11:04:05	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHERE Account...	0 row(s) affected	0.047 sec
8	11:04:36	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
9	11:06:03	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Customer_ID, C.Custom...	0 row(s) affected	0.047 sec
10	11:06:16	CALL GetCustomerAccounts(101)	1 row(s) returned	0.031 sec / 0.000 sec
11	11:07:09	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account ...	0 row(s) affected	0.047 sec
12	11:07:13	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.016 sec

15. PROCEDURE TO WITHDRAW MONEY

```
186
187 •   CREATE PROCEDURE WithdrawMoney(
188         IN p_Account_No INT,
189         IN p_Amount DECIMAL(12,2)
190     )
191     BEGIN
192
193         UPDATE Account
194         SET Balance = Balance - p_Amount
195         WHERE Account_No = p_Account_No;
196
197     END //
198
199     DELIMITER ;
200
201 •   CALL WithdrawMoney(10001, 3000);
202
203
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	11:04:36	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
9	11:06:03	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Customer_ID, C.Custom...	0 row(s) affected	0.047 sec
10	11:06:16	CALL GetCustomerAccounts(101)	1 row(s) returned	0.031 sec / 0.000 sec
11	11:07:09	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account ...	0 row(s) affected	0.047 sec
12	11:07:13	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.016 sec
13	11:07:46	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account...	0 row(s) affected	0.047 sec
14	11:07:51	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.047 sec

16. TRIGGER – PREVENT NEGATIVE BALANCE

```
211
212     IF NEW.Balance < 0 THEN
213
214         SIGNAL SQLSTATE '45000'
215         SET MESSAGE_TEXT =
216         'Transaction failed: Insufficient balance';
217
218     END IF;
219
220 END //
221
222 DELIMITER ;
223
224
225
226 • UPDATE Account
227   SET Balance = Balance - 60000
228   WHERE Account_No = 10001
229
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	11:06:16	CALL GetCustomerAccounts(101)	1 row(s) returned	0.031 sec / 0.000 sec
11	11:07:09	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account ...	0 row(s) affected	0.047 sec
12	11:07:13	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.016 sec
13	11:07:46	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account...	0 row(s) affected	0.047 sec
14	11:07:51	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.047 sec
15	11:09:10	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Balance < 0 THEN ...	0 row(s) affected	0.031 sec

```
224
225
226 •  UPDATE Account
227    SET Balance = Balance - 60000
228    WHERE Account_No = 10001
229
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	11:06:16	CALL GetCustomerAccounts(101)	1 row(s) returned	0.031 sec / 0.000 sec
11	11:07:09	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account ...	0 row(s) affected	0.047 sec
12	11:07:13	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.016 sec
13	11:07:46	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account...	0 row(s) affected	0.047 sec
14	11:07:51	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.047 sec
15	11:09:10	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Balance < 0 THEN ...	0 row(s) affected	0.031 sec
16	11:09:14	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.046 sec

17. TRIGGER – PREVENT NEGATIVE DEPOSIT

```
231 •  CREATE TRIGGER CheckTransactionAmount
232    BEFORE INSERT ON Bank_Transaction
233    FOR EACH ROW
234    BEGIN
235        IF NEW.Amount <= 0 THEN
236            SIGNAL SQLSTATE '45000'
237            SET MESSAGE_TEXT =
238            'Transaction amount must be greater than zero';
239        END IF;
240    END //
241
242    DELIMITER ;
243 •  INSERT INTO Bank_Transaction
244    (Account_No, Transaction_Type, Amount)
245    VALUES
246    (10001, 'DEPOSIT', -5000);
247
248 •  INSERT INTO Bank_Transaction
249    (Account_No, Transaction_Type, Amount)
250    VALUES
251    (10001, 'DEPOSIT', 5000);
252
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	11:07:46	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDATE Account...	0 row(s) affected	0.047 sec
14	11:07:51	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.047 sec
15	11:09:10	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Balance < 0 THEN ...	0 row(s) affected	0.031 sec
16	11:09:14	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.046 sec
17	11:12:07	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BEGIN IF NEW.Amou...	0 row(s) affected	0.016 sec
18	11:12:26	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.047 sec
19	11:12:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.047 sec

18. CREATE TRANSACTION AUDIT TABLE.

```
252
253 •  CREATE TABLE Transaction_Audit (
254        Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
255        Transaction_ID INT,
256        Account_No INT,
257        Transaction_Type VARCHAR(20),
258        Amount DECIMAL(12,2),
259        Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
260    );
261
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	11:07:51	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.047 sec
15	11:09:10	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Balance < 0 THEN ...	0 row(s) affected	0.031 sec
16	11:09:14	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.046 sec
17	11:12:07	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BEGIN IF NEW.Amou...	0 row(s) affected	0.016 sec
18	11:12:26	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.047 sec
19	11:12:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.047 sec
20	11:13:29	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID INT, Account_No ...	0 row(s) affected	0.078 sec

19. TRIGGER – TRANSACTION AUDIT

```
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN

    INSERT INTO Transaction_Audit
    (
        Transaction_ID,
        Account_No,
        Transaction_Type,
        Amount
    )
    VALUES
    (
        NEW.Transaction_ID,
        NEW.Account_No,
        NEW.Transaction_Type,
        NEW.Amount
    );

END //

DELIMITER ;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	11:09:10	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Balance < 0 THEN ...	0 row(s) affected	0.031 sec
16	11:09:14	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.046 sec
17	11:12:07	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BEGIN IF NEW.Amou...	0 row(s) affected	0.016 sec
18	11:12:26	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.047 sec
19	11:12:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.047 sec
20	11:13:29	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID INT, Account_No ...	0 row(s) affected	0.078 sec
21	11:14:34	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INSERT INTO Transacti...	0 row(s) affected	0.015 sec

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);


SELECT * FROM Transaction_Audit;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
1	7	10001	DEPOSIT	2500.00	2026-08-23 11:14:52
NULL	NULL	NULL	NULL	NULL	NULL

Transaction_Audit 11

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	11:12:07	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BEGIN IF NEW.Amou...	0 row(s) affected	0.016 sec
18	11:12:26	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.047 sec
19	11:12:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.047 sec
20	11:13:29	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID INT, Account_No ...	0 row(s) affected	0.078 sec
21	11:14:34	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INSERT INTO Transacti...	0 row(s) affected	0.015 sec
22	11:14:52	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 2500)	1 row(s) affected	0.016 sec
23	11:14:56	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.015 sec

20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 5000);


SELECT *
FROM Account
WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	52000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 12

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	11:12:26	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.047 sec
19	11:12:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.047 sec
20	11:13:29	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID INT, Account_No ...	0 row(s) affected	0.078 sec
21	11:14:34	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INSERT INTO Transacti...	0 row(s) affected	0.015 sec
22	11:14:52	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 2500)	1 row(s) affected	0.016 sec
23	11:14:56	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.015 sec
24	11:15:57	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec